Anybody who watches too many sci-fi movies knows that gamma rays are bad for you. You might turn green and muscular, or spiderwebs might squirt from your wrists. But they're also hard to trap. They pass right through ordinary lenses and mirrors. How, then, to observe them? The guts of *Explorer XI*'s telescope held a device called a scintillator, which responds to incoming gamma rays by pumping out electrically charged particles. If you measure energies of the particles, you can tell what kind of high-energy light created them.

Two years later the Soviet Union, the United Kingdom, and the United States signed the Limited Test Ban Treaty, which prohibited nuclear testing underwater, in the atmosphere, and in space—where nuclear fallout could spread and contaminate places outside your own country's perimeter. But this was the Cold War, a time when nobody believed anybody about anything. Invoking the military edict "trust but verify," the U.S. deployed a new series of satellites, the *Velas*, to scan for gamma ray bursts that would result from Soviet nuclear tests. The satellites indeed found bursts of gamma rays, almost daily. But Russia wasn't to blame. These came from deep space—and were later shown to be the calling card of intermittent, distant, titanic stellar explosions across the universe, signaling the birth of gamma ray astrophysics, a new branch of study in my field.

In 1994, NASA's Compton Gamma Ray Observatory detected something as unexpected as the *Velas*' discoveries: frequent flashes of gamma rays right near Earth's surface. They were sensibly dubbed "terrestrial gamma-ray flashes." Nuclear holocaust? No, as is evident from the fact that you're reading this sentence. Not all bursts of gamma rays are equally lethal, nor are they all of cosmic origin. In this case, at least fifty bursts of these flashes emanate daily near the tops of thunderclouds, a split second before ordinary lightning bolts strike. Their origin remains a bit of a mystery, but the best explanation holds that in the electrical storm, free electrons accelerate to near the speed of light and then slam into the nuclei of atmospheric atoms, generating gamma rays.

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Today, telescopes operate in every invisible part of the spectrum, some from the ground but most from space, where a telescope's view is unimpeded by Earth's absorptive atmosphere. We can now observe phenomena ranging from low-frequency radio waves a dozen meters